



1721 - Probing the Interstellar Medium of Galaxies in the Early Universe

Cycle: 1, Proposal Category: AR

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OBSERVATIONS

ABSTRACT

Upon the launch of the James Webb Space Telescope, we will have the ability to probe rest-frame optical and UV emission line diagnostics from cosmic noon at $z \sim 2$ out to the epoch of reionization, free from the interference of atmospheric sky lines that have hampered observations from the ground. The Cosmic Evolution Early Release Science (CEERS) survey will provide the extragalactic community with the first opportunity to probe fundamental parameters of the interstellar medium (ISM) in a large sample of high redshift galaxies. CEERS will conduct a high redshift spectroscopic survey, including the first large spectroscopic survey of the reionization era, and will open up new optical diagnostics for ~ 300 high redshift galaxies, including ~ 30 targets at $z > 6$. With wavelength coverage of 1-5 microns, CEERS will probe [OIII] out to $z = 9$ and H α to $z = 6.6$, while also tracing valuable UV diagnostic lines such as [CIII] and [CIV] at the epoch of reionization. Using the full sample of CEERS star-forming galaxies, we propose to trace the evolution of metallicity, ionization parameter, and gas pressure over the history of the universe and reveal their relationships with galaxy stellar mass, star formation rate (SFR), redshift. This work will give insight into the relationship between star formation and the ISM in the early universe and the evolution of ISM properties across cosmic time.

OBSERVING DESCRIPTION

There are no new observations requested as part of this archival proposal.

This proposal will make use of the publicly available NIRSpec spectroscopy of high redshift galaxies taken as a part of the CEERS (Cosmic Evolution Early Release Science) survey (PID 1345).